Technical Report: Periodic-1D Higher-Rank Composite-Band Alignment and Locality

Project Koios

Pre-execution architecture and mathematical contract

1 Scope and evidence boundary

This report documents the finite mathematics, numerical conventions, and software-owner boundary of the periodic one-dimensional higher-rank controlled calculation used by the Paper C development program. It is calculation-owned and publication-independent. The selected protocol remains the authority for coefficients, controls, tolerances, schemas, and future retained evidence. This report explains that contract; it does not replace it.

The calculation-local `workflow/` package and its ignored summaries are replaceable development material. They are not a maintained runtime, retained run, convergence result, or evidence package. No result reported below was obtained from a new execution. The labels D1, P0, and G0 are retained only where they identify the frozen protocol. They are not proposed public software class names.

2 Finite model inventory

The selected ladder uses retained ranks

$$r \in \{3, 4, 8, 16\}, \qquad L \in \{1, 2, 4, 8\},$$

ambient dimension $2r$, construction meshes $M \in \{513, 1025, 2049\}$, an independent evaluation mesh of size 8191, and hopping supports $S \in \{0, 1, 2, 4, 8, 16, 32, 64, 128\}$. All construction meshes are odd. Their complete centered translation inventories are therefore ordinary inversion-complete sets with no self-inverse Nyquist representative.

Four alignment controls remain distinct: no alignment, one global unitary, pointwise complete unitaries, and the known generating inverse. The known inverse is an oracle control and is not evidence of practical recoverability. Transport of the analytic reference frame precedes these alignment controls and is shared by them.

3 State spaces and action conventions

At each reciprocal coordinate, an ambient frame is a matrix

$$F(k) \in \mathbb{C}^{2r \times r}, \qquad F(k)^\dagger F(k) = I_r.$$

Its columns use one declared ambient basis and span a rank-r retained space. A retained coordinate transformation acts on the right,

$$F(k) \longmapsto F(k)U(k), \qquad U(k) \in U(r),$$

whereas the reciprocal-endpoint sewing map acts on the left in the ambient space,

$$F(-\pi) \longmapsto S_G F(-\pi), \qquad S_G \in \mathbb{C}^{2r\times 2r}.$$

For the selected P0 construction, $S_G = I_{2r}$, but the route still records and verifies that declaration. Identity sewing is not inferred from periodic eigenvalues or array shape.

If $H_p(k)$ is the parent and $H_F(k) = F(k)^\dagger H_p(k) F(k)$ is its represented Hamiltonian, then a right frame action gives the covariant matrix transformation

$$H_{FU}(k) = U(k)^\dagger H_F(k) U(k).$$

Transport, alignment, and Fourier transformation therefore have ordered and generally noncommuting roles.

4 Sequential polar transport

For one odd half-open mesh, let

$$k_n = -\pi + \frac{2\pi n}{M}, \qquad n = 0, \ldots, M-1,$$

and append the direct endpoint $k_M = +\pi$. Write the analytically sampled frame as $Y_n = F(k_n)$, including $Y_M = F(+\pi)$. Set $\widetilde{F}_0 = Y_0$. Given the transported frame at index n, form

$$O_n = \widetilde{F}_n^\dagger Y_{n+1} = L_n \Sigma_n R_n^\dagger, \qquad Q_n = R_n L_n^\dagger, \qquad \widetilde{F}_{n+1} = Y_{n+1} Q_n. \tag{1}$$

This is the right-unitary Procrustes transformation that aligns the next raw frame toward the already transported predecessor. Indeed,

$$\widetilde{F}_n^\dagger \widetilde{F}_{n+1} = L_n \Sigma_n L_n^\dagger,$$

which is Hermitian positive semidefinite.

The complete singular spectrum is retained for every attempted link. With

$$\tau_{\mathrm{sv}}(O_n) = \max\!\left(10^{-12}, 10^{-10}\sigma_{\max}(O_n)\right),$$

the link qualifies only when $\sigma_{\min}(O_n) > \tau_{\mathrm{sv}}(O_n)$. Equality is deficient. If a neighbor link is deficient, no null-space completion is applied: that link is recorded as deficient and all links that depend on its transported output are recorded as unattempted. A backend decomposition failure is separate from mathematical rank deficiency and carries a stable operation, reason, and link index.

5 Endpoint mismatch, holonomy, and distributed closure

After the endpoint step, define the protocol-oriented endpoint mismatch

$$S = \widetilde{F}_0^\dagger \widetilde{F}_M. \tag{2}$$

Let its admitted unitary polar factor be

$$\mathcal{H} = Z\,\mathrm{diag}(e^{i\gamma_1}, \ldots, e^{i\gamma_r}) Z^\dagger, \qquad \gamma_a \in (-\pi, \pi]. \tag{3}$$

The branch is part of the method. At eigenvalue -1, this convention chooses $\gamma = +\pi$, not $-\pi$. Both signs exponentiate to the same endpoint eigenvalue, but they select different fractional matrix roots

and therefore different intermediate gauges. The branch cannot be chosen after observing a locality outcome.

The periodically corrected frame is

$$F_n^{\mathrm{pt}} = \widetilde{F}_n Z \,\mathrm{diag}\Big(e^{-i(n/M)\gamma_1}, \ldots, e^{-i(n/M)\gamma_r}\Big)\, Z^\dagger, \qquad n = 0, \ldots, M. \tag{4}$$

The phases are stored deterministically for reproducibility, but their scientific meaning is an unordered multiset. Sorting is a storage convention, not a band labeling through degeneracies.

Proposition 1 (Finite endpoint closure)**.** *If $\widetilde{F}_M = \widetilde{F}_0\mathcal{H}$ and Equations (3)–(4) hold, then $F_M^{\mathrm{pt}} = F_0^{\mathrm{pt}}$.*

Proof. At $n = M$, the correction is $Z\,\mathrm{diag}(e^{-i\gamma_a})Z^\dagger = \mathcal{H}^\dagger$. Hence

$$F_M^{\mathrm{pt}} = \widetilde{F}_M\mathcal{H}^\dagger = \widetilde{F}_0\mathcal{H}\mathcal{H}^\dagger = \widetilde{F}_0 = F_0^{\mathrm{pt}}.$$

□

For the selected P0 family, the analytic endpoint obeys $Y_M = S_G Y_0$ exactly. In that setting the Optimization owner retains both orientations. Its closure right action $\mathcal{A}$ aligns the last primitive frame toward the sewn first frame,

$$\widetilde{F}_{M-1}\mathcal{A} \approx S_G\widetilde{F}_0,$$

and its endpoint mismatch is $\mathcal{E} = \mathcal{A}^\dagger$, matching the protocol orientation of $\mathcal{H}$. Schur phases are phases of $\mathcal{E}$, not $\mathcal{A}$. The owner correction step is

$$\mathcal{W} = Z\,\mathrm{diag}(e^{-i\gamma_a/M})Z^\dagger, \qquad \mathcal{W}^M = \mathcal{E}^\dagger = \mathcal{A},$$

and sample j receives the right action $\mathcal{W}^j$. The explicit owner branch `negative_pi_open`, whose value is $(-\pi, \pi]$, therefore reproduces the protocol directly, including $\gamma = +\pi$ and $\mathcal{W} = e^{-i\pi/M}$ at eigenvalue -1. An interface that stored phases of $\mathcal{A}$ instead would require the inverse branch $[-\pi, \pi)$ to reproduce the same path.

Remark 1. *A Schur basis inside an exactly degenerate phase subspace need not be unique. The matrix function is nevertheless invariant under basis changes that remain within a subspace assigned the same branch value. Near degeneracy, backend-specific Schur vectors are numerical evidence, not canonical physical band labels.*

6 Metric-explicit diagnostics

All scientific diagnostics name their norm and normalization. For an r-column frame or an $r \times r$ unitary, the protocol-normalized Frobenius defect is

$$d_{F,\mathrm{norm}}(A, B) = \frac{\|A - B\|_F}{\sqrt{r}}, \qquad d_{U,\mathrm{norm}}(U) = \frac{\|U^\dagger U - I_r\|_F}{\sqrt{r}}.$$

The reciprocal endpoint checks are

$$d_{\mathrm{frame,endpoint}} = \frac{\|F(+\pi) - F(-\pi)\|_F}{\sqrt{r}},$$

$$d_{\mathrm{projector,endpoint}} = \frac{\|P(+\pi) - P(-\pi)\|_F}{\sqrt{r}}, \qquad P = FF^\dagger,$$

and the primitive closure check is

$$d_{\text{closure}} = \frac{\|F_M^{\text{pt}} - F_0^{\text{pt}}\|_F}{\sqrt{r}}.$$

The selected endpoint and primitive-closure tolerance is 10^{-11}. The P0 closure phase gate is

$$d_{\text{phase}} = \max_a |\gamma_a| \leq 10^{-10} \text{ radian}.$$

This is a protocol integrity check for the exactly trivial control, not a generic demand that physical holonomy vanish.

A reusable numerical owner may additionally retain a maximum-absolute-entry defect for self-validation. Such a value is not interchangeable with a Frobenius defect and must carry a metric-explicit name. KSD computes and retains the protocol diagnostics through the canonical norm owners rather than relabeling an owner-internal validation metric.

Table 1 is a calculation-facing crosswalk of the frozen gates. The protocol remains authoritative if this explanatory table and the selected version ever differ.

Table 1: Frozen P0 higher-rank numerical tolerances.

Diagnostic	Normalization	Tolerance
momentum and endpoint coordinates	radian	10^{-14}
frame orthonormality, gauge/alignment unitarity, projector idempotency	raw Frobenius	10^{-12}
Procrustes stationarity, objective identity, and negative polar part	dimensionless	10^{-12}
smooth/attacked projector equality	raw Frobenius	10^{-12}
direct ordered-spectrum equivalence	E_{ref}	10^{-11}
A2/A3 frame recovery	$\sqrt{r}$	10^{-11}
A2/A3 represented-operator recovery	$\sqrt{r}E_{\text{ref}}$	10^{-11}
endpoint sewing and primitive closure	normalized Frobenius	10^{-11}
maximum closure-phase magnitude	radian	10^{-10}
Fourier reconstruction and Hermitian pairing	$\sqrt{r}E_{\text{ref}}$	2×10^{-11}
relative Parseval defect	dimensionless	2×10^{-11}
error decomposition and support monotonicity	dimensionless	2×10^{-11}
maximum perturbation-bound violation	E_{ref}	10^{-11}
final interpolant operator error	$\sqrt{r}E_{\text{ref}}$	10^{-9}
final interpolant ordered-spectrum maximum error	E_{ref}	10^{-10}
retained-block refinement, $\lvert R\rvert \leq 128$	$\sqrt{r}E_{\text{ref}}$ per block	10^{-10}

7 Alignment and represented operators

Transport chooses a coherent reference frame. It does not by itself align an attacked candidate. The four protocol channels are subsequently applied to the candidate from the right. Pointwise and one-global Procrustes alignment solve different optimization problems; the known inverse is an oracle; identity alignment performs no fit. A rank-deficient global aggregate makes only that restricted channel inapplicable. A rank-deficient pointwise overlap invalidates the selected P0 case because the analytic construction predicts full rank.

The KSD disposition inventory is closed:

- `qualified` means every required rank, unitary, stationarity, objective, positive-semidefinite, oracle, and independent-comparison gate passes;
- `aggregate-rank-deficient` is a bounded A1 result: no A1 transformation or downstream A1 locality value is emitted;
- `pointwise-rank-deficient` invalidates the complete P0 case;
- `alignment-numerically-unresolved` records nonfinite input or typed backend failure without emitting a transformation;
- `alignment-integrity-failed` records a nominally full-rank A1/A2 factor that fails a required integrity or independent-comparison gate; and
- `oracle-integrity-failed` records failure of A3 analytic unitarity, frame recovery, operator recovery, or independent reconstruction.

Stable backend codes distinguish nonfinite arithmetic, nonconvergence, and validation failure; traceback text and host paths are not scientific fields.

Every admitted action $U_A(k)$ transforms the represented Hamiltonian before Fourier conversion,

$$H_A(k) = U_A(k)^\dagger H_C(k) U_A(k).$$

Aligning already truncated hopping blocks is a different operation and is not substituted for reciprocal-space alignment.

8 Odd-mesh Fourier representation and locality

For odd M, the complete ordinary translation set is

$$\mathcal{R}_M = \left\{ -\frac{M-1}{2}, \dots, \frac{M-1}{2} \right\}.$$

For each applicable channel,

$$T_{A,M}(R) = \frac{1}{M} \sum_{n=0}^{M-1} e^{-ik_n R} H_A(k_n), \qquad H_{A,M}(k) = \sum_{R \in \mathcal{R}_M} e^{ikR} T_{A,M}(R).$$

There is no Nyquist singleton and no cyclic-to-ordinary removal channel. Support selection is a separate KSD policy: $\Omega_S = \{R : |R| \le S\}$. The complete transform, selected support, and deliberate truncation remain separate records.

On the withheld mesh, the operator residuals are

$$e_{\text{interp}} = H_{A,M} - H_A^{\text{direct}}, \qquad e^{(S)}_{\text{trunc}} = H^{(S)}_{A,M} - H_{A,M},$$
$$e^{(S)}_{\text{total}} = H^{(S)}_{A,M} - H_A^{\text{direct}} = e_{\text{interp}} + e^{(S)}_{\text{trunc}}.$$

Only the pointwise residuals telescope. Their scalar maxima are not added, and a cancellation in the total channel is reported rather than called convergence. For each channel $x \in \{\text{interp}, \text{trunc}, \text{total}\}$, cross-rank operator summaries use, for example,

$$\epsilon^{(x)}_{F,\max} = \max_k \frac{\|e_x(k)\|_F}{\sqrt{r} E_{\text{ref}}}, \qquad \epsilon^{(x)}_{F,\text{RMS}} = \left(\frac{1}{N_{\text{eval}}} \sum_k \frac{\|e_x(k)\|_F^2}{r E_{\text{ref}}^2} \right)^{1/2}.$$

Ordered-spectrum errors are normalized by E_{ref}. These are separate from spectral-norm perturbation evidence and from hopping-tail weights.

9 Software-owner boundary

The intended dependency direction is

$$\texttt{ksdft2effmass} \longrightarrow \texttt{projectkoios.physkit} \longrightarrow \texttt{projectkoios-optimization}.$$

Optimization owns unit-free right-unitary Procrustes transport, numerical-rank evidence, closure decomposition, fractional-root construction, and typed backend failures. Its current contract requires an explicit `ClosedLoopPolarFrameTransportPrincipalPhaseBranch`; KSD selects `negative_pi_open`. The result separately retains `closure_alignment_right_action`, `closure_endpoint_mismatch`, `distributed_closure_correction_step`, and the deterministically stored endpoint-mismatch phase tuple. Owner validation defects explicitly name their maximum-absolute-entry metric.

PhysKit binds those arrays to the ordered reciprocal route or one-dimensional finite-periodic mesh, ambient-basis identity, retained-frame identity, and endpoint-sewing semantics. KSD owns the synthetic parent, scientific tolerances, normalized Frobenius diagnostics, alignment-channel dispositions, Fourier support policy, result wire, independent verification, and claim resolution.

The Optimization implementation is committed at `ef608716ae9f515f56406dd517c1a8b7487d3391` and published on its work branch. The adapted PhysKit semantic wrapper is locally committed at `3371d8883c85fdd16754cffb29d3f4ec21dd7861` but has not been pushed. Neither owner revision is yet integrated through KSD's declared dependencies. This report therefore records the target boundary without claiming that the maintained KSD runtime already consumes it. The calculation-local `workflow/` package must not be copied or imported into the runtime.

10 Independent verification boundary

The future producer uses canonical owner actions. The independent reference analyzer must reconstruct the finite formulas without calling the producer analyzer. At minimum it independently checks:

1. ordered point and sewing identities;
2. every overlap spectrum, cutoff, strict margin, and rank disposition;
3. right-action orientation and transported-frame reconstruction;
4. endpoint mismatch or inverse-closure orientation;
5. Schur reconstruction, selected phase branch, fractional root, and root power;
6. endpoint and primitive-closure normalized Frobenius defects;
7. represented-operator covariance;
8. direct Fourier sums, Hermitian pairs, Parseval, support masks, and tail norms; and
9. interpolation, truncation, total-error, and perturbation-bound channels.

One byte-owning verifier admits the exact immutable result bytes, composes this reference analysis, and reports a closed inventory of scientific comparisons. Passing establishes bounded software and finite-protocol numerical behavior only.

11 Limitations and nonclaims

The selected family is a topologically trivial synthetic control. Closure phases are transport and sewing diagnostics, not a Chern, polarization, or Wannier-center result. The distributed closure is a smooth-gauge construction, not a localization optimum. No two- or three-dimensional higher-rank calculation exists. Moving retained subspaces, complex gauge phases, restricted-circuit fitting, patchwise frames, and material specialization remain prospective. Current work establishes neither continuum convergence nor scientific validation.

12 Next gate

The next KSD gate is a stable typed request/result contract built on integrated owner dependencies, followed by a producer-independent direct-formula analyzer and authenticated rendering. Only after implementation review, strict schema validation, complete tests, and an unreleased resource-measured dry run may a separately authorized retained execution be considered.